Unit 30 - Digital Graphics
Assignment 1 - Graphics Hardware and Software

Introduction

Scenario / Vocational Context: *You have been employed as a trainee at SuperSkills Studios. The company would like you to spec up a PC for you to use during your employment and to justify why the hardware and software you have chosen will aid you to create high-end graphics. You will then have to explain your understanding of the limitations of different hardware and software as well as detail your thoughts on how evolving output mediums have impacted graphic design and creation.*

Task 1

(U30:P1) - Describe the hardware and software used to create and edit graphic images

Task description: *You are to describe what each component does/why it is needed as well as why the one you have chosen will help you in creating, and working with, high-end graphics.*

Processor + RAM

The processor, or CPU, is one of the most important parts of a computer system, because it is considered the “brain” of the computer. What I mean by that is that the processor caches and moves a specific amount of data that is found encoded within most or all of the computer programs. The processor takes all the information that has been gathered within the program, and interprets it from a series of numbers and symbols, into a language in which humans can understand.

The best choice for a processor is probably one of the Intel “Core i” processors, especially the ones you can best afford. The software will run with a Pentium or Celeron from Intel, but not as well as the Core i processors. That is the reason I have picked the *Intel Core i7-7700k Processor (8M cache, up to 4.50GHz)* as the processor for this build.

The performance of this processor has a base frequency of 4.20 GHz and a max turbo frequency of 4.50 GHz, it also has a Bus Speed of 8 GT/s. This means that the processor is extremely good at handling at multitasking very big software such as Adobe Photoshop with its processing speed. The bus speed is helpful because it determines how long it would take the data to launch the software to process.

The graphics specification of the CPU is that it has 4K support at 60Hz and it has a Max resolution of 4096 x 2404@14Hz for HDMI 1.4 and 4096 x 2404@60Hz at DP and eDP. This is useful because it can handle such a high resolution, which is useful for graphics design because high quality content can be produced which is very desirable when people look at graphics design.

The reason a processor impact graphical work is because it executes instructions which are

heavily associated with graphics. For example, loading graphics from the memory or storage, rendering images, applying effects to an image and converting the file types of an image. In order for these operations to work fast and efficiently, it would need to require quite a high amount of processing speed, which is why that the processor that has been picked has a clock speed of up to 4.50 GHz.

The processing speed of the CPU also has an impact with the Graphics Processing Unit (the GPU). The reason for this is due to the fact that the speed in which the information is being processed isn't very quick, making the graphics processor too long to produce the information into a graphic.

A CPU also has multiple cores instead of one single core. The reason for this is so the processor provides an efficient amount of processing speed and also giving the ability to multitask, since many graphics software is very demanding for the components of the PC.

Random Access Memory, or RAM, is a piece of hardware that can be used to read and write data while the computer is turned on and contains memory storage that is volatile. THis means that contents are stored within RAM only while the PC is tuned on, the memory will be available on the RAM until the PC is shut down. The main purpose for RAM is to load programs from the place it is stored in (ROM) into the CPU, or processor.

The access time for each application load into the RAM will be the same due to the memory being random access type. However, the rate at which a program load will have an effect on how quick the program will run. If the RAM cannot load a program because it is full, the process will load the program directly from ROM, which is a much slower process.

Since RAM is a piece of hardware that can be replaced and added to a computer, more RAM can be added to a PC to improve its performance. Especially due to the fact the average motherboard has 4 RAM slots and it is usually occupied by 2 pieces of RAM.

RAM is probably the most important component when talking about graphics design. The reason for this is because graphics software such as Adobe’s software are memory hogs. Its best to go with 32GB of RAM because the software would never hit the RAM’s limit. 16GB of RAM could handle the softwares pretty well, and 8GB you can get by with.

However, 8GB should be the bare minimum of RAM for graphics design. That is the reason why I chose the *Corsair Dominator Platinum Series 32GB (4 x 8GB) DDR4 DRAM 3200MHz C16 Memory Kit* RAM for this build.

The memory technology that is in the RAM is that it is a DDR4. The reason I picked DDR4 is because it is the latest in graphics card technology and it has a much higher speed. Which, of course, is much better for graphics design because it can handle graphics software a lot better and quicker than a DDR3.

For graphics design, RAM loads graphics such as images, videos and even the software in which it contains the graphics, which then transfers data to the graphics card’s memory, VRAM, to display it on the monitor screen. If there was a lack of VRAM space, the program/graphic will take the information directly from the RAM, which is a significantly slower process. The

gigabytes of RAM and the type of DDR it is in the PC will determine how much more performance the computer has and also the ability to run multiple programs/graphics at the same time.

Graphics Card

A graphics card, otherwise called a video card or GPU, is a piece of hardware that send information from the computer to the monitor, which will display the image on the monitor screen. Difference graphics card have different and complicated tasks. For example, a graphics card can operate one output to multiple monitors, display different outputs to different monitors or just watching TV from the computer. These features is what makes the graphics cards one of the most versatile component within a PC.

With graphics design, the best is to go with Nvidia graphics. It isn't because AMD graphics are bad, but it's because most graphics software such as Adobe doesn't support anything but Nvidia very well. Hence the reason that the graphics card that I have picked for this build is the *Nvidia PNY Quadro P6000 24GB GDDR5X Professional Graphics Card.*

The graphics card has an output Interface of 1x DVI-D, 4x DisplayPort, 3x DisplayPort to DVI-D Adapter and 1x Stereo Adapter. It had a memory type of 384-bit GDDR5X and a memory size of 24 GB. It has a multi monitor support of up to 4 Displays, a max DisplayPort resolution of 7680 x 4320 and a max DVI-D resolution of 2560 x 1600.

These are all very good for graphics design because of the insane power this GPU holds, Nvidia has proudly written this graphics card as "*The NVIDIA Quadro P6000 is the world's most advanced professional graphics solution ever created, combining the latest GPU, memory and display technologies that result in unprecedented performance and breakthrough capabilities*".

Graphics cards have a big impact with graphic designing, the reason being is that they are needed to display the content on the monitor and to process changes in the graphic displayed on-screen. Having a high end graphics card will give many benefits towards graphic design work because the computer would be able to load graphics a lot faster and have a much faster response time to what the user does within a software.

Modern graphics cards have become much more efficient with RAM, using less of the memory than before, making the system perform better than previously. This allows applications such as Adobe's Photoshop to run more efficiently.

Lastly, a decent graphics card with a DVI connection will be able to use "truecolour", which is basically the ability to have 32-bit colour depth. What I mean by this is that the colours that will be displayed on the monitor screen will consist of 16.7 million different colours, which is a much more significant amount to 26-bit (which has 65,536 colours) and 8 bit (which has 256 colours). This makes software packages, such as Photoshop, to use the much wider colour variety to produce the image within the software, giving it a much more realistic colour depth.

SD Card Reader

An SD card reader is a piece of hardware that is used to access the image files that are stored in an SD card. With modern laptops and very rarely desktops, there are already built-in SD cards. However, if there isn't a pre-installed SD card, a USB connected SD card reader can be used instead, there is no compatibility difference between an SD card that is pre-installed or connected via USB.

With graphics design, it is important how fast the transfer between the SD card and the PC is, the reason for this is due to the fact that modern cameras are getting much higher DPI or “MegaPixels”. This is the reason I decided to get a high end, high speed SD card reader like the *02 Transcend USB 3.0 Card Reader RDF8.*

I picked this SD card reader for this build because it has a fast read and write time of 130MB/s read and 95MB/s write. It also has a USB 3.1/3.0 port, which is significantly faster than the previous USB 2.0

Even though SD card readers are mainly used for an SD card, which is the most common form of storage due to it being compatible with any digital camera, there are several different types of memory card which can vary in size, price, storage and connection type.

A few SD cards can have up to 128GB of storage which is a very good size even for high quality images from a camera such as .RAW image files which are usually large in file size, especially for high megapixel cameras.

There are also version of digital card storage such as: SDHC, which is identical to an SD card but on average has double the storage capacity; CF cards, which is the other type of memory card that can also be stored within cameras, it is similar to an SD card but often larger than a traditional SD card.

SD card readers can be very useful for a graphics designer and they can really impact the benefits and drawbacks of a graphics designer for image or any other file transfer onto a PC.

The benefits it has is that it can be used to transfer images from a digital camera onto a computer via SD card reader than it is with other common methods of transfer such as a USB connection to a camera. An SD card reader also allows you to transfer the same original image to other PCs with a very compact and fast methods of transferring.

The drawbacks that an SD card reader could have is that it is possible that the files within an SD card have a chance of being corrupted, which could make you lose the uncompressed, original image or the only image from the user.

Also, since an SD card reader does not support the other types of digital cards such as SDHC and CF cards, it would mean the the SD card reader that has been purchased would be obsolete, and forcing the other forms of digital cards that aren't SD cards could cause severe or irreversible damage to the SD card reader.

Hard Drive

The hard drive is the main form of storage for a computer and it allows for fast random access to a large amount of data. With modern hard drives getting much larger in size, it makes it much more common to find a hard drive that contains storage that is in at least 2TB (TeraBytes). Hard Drives are able to store and retrieve data much faster than other methods of secondary and mobile storage, such as CD-ROM and USB flash drives.

Since the hard drive is the main storage for a computer, that means that it will contain the graphics that the user would have transferred into the hard drive and even the graphics software such as Adobe Photoshop, GIMP and MicroSoft Paint.

When the software is being used, the hard drive is the top priority in which the computer will search for it, so that the files that are stored within the hard drive will be found almost immediately, the same goes for opening and saving the software.

THe hard drive is usually stored within the PC itself, however, it is possible to have an external hard drive connected to the PC. This has its benefits because there is a significant amount of space to contain the hard drive, instead of the close space of a PC case. It also gives the ability to have a lot more hard drives due to the significant amount of space.

There are mainly two types of hard drive storage: Hard Disk Drive and Solid State Drive. The Hard Disk Drive contains a magnetic disk which spins at a very fast RPM and it has a pin which is used to fetch the data within the magnetic disk. The Solid State Drive, or SSD, uses flash memory technology, similar to the memory on the phone and the SD cards for a camera.

Solid State Drives are much faster at reading and writing than the traditional magnetic spinning disks, which make a huge difference for graphics editing. This means that the SSD hard drive is the best choice for storage for the PC build. That is the reason I decided to include the Samsung 850 EVO 4 TB Internal SSD - 2.5" - MZ-75E4T0 - SATA as the storage unit for this build.

The reason I chose this Hard drive for the build is because it is an SSD drive which means it is a lot faster than a HDD as I mentioned before. It also have 4TB of storage space so it can last a long time with digital graphics due to its colossal storage space.

1 Additional Input Device (e.g. camera, tablet)

A digital camera is a device that has the ability to “capture” a real life object into a digital representation of that object. IT does this by using light to capture and manipulates the light in order for the image to be constructed identically with the real life object.

Modern digital cameras don't use film like older cameras, instead they use a CMOS or CCD image sensor to sense the different levels of light across the plane and stores the image in a flash memory device which is the SD card. The captured images on the digital camera can be transferred onto a computer either by a USB cable or by removing the SD card and inserting it into an SD card reader.

For graphics designers, they have to choose throughout a wide range of digital cameras, such

as cost (which is unimportant due to no budget limit on this build), the type of lens the camera has, and most importantly the amount of megapixels the camera has.

The type of lens has an impact on the quality of the image that has been captured. The reason for this is due to the optical zoom that the lens have, which allows the user to choose how much of the image can be captured and appear within the graphic image.

Likewise, the certain amount of megapixels is needed when using a digital camera for graphics design as it determines how many pixels are captured within a single image. The higher the amount of pixels, the better quality the image is, which allows for better image editing and manipulation via software.

This is important due to the ability to crop an image and still keep the image a good quality without pixilation or if a high quality image is needed due to the image being printed off or transferred as a large image. This is the reason I bought the *Sony Alpha A7R II*.

The use of the digital camera has an effect with graphics design because the camera can be used to capture the desired image or reference material. Having a good camera with a large amount of megapixels which gives the ability to take much more clear and sharper images so they can be edited via a software so that a much better end result will be produced.

Monitor

The monitor or Visual Display Unit displays the results of the computer's processing activity using the information from the graphics card, the displayed information within the monitor screen allows the user to interact within the computer and the systems.

With monitors, there are two different methods of display: CRT (Cathode Ray Tube) and LCD (Liquid Crystal Display). CRT monitors are very uncommon to find in the modern day because it was used between the 1960's and 2000's, which was later replaced by the LCD screen. The reason it has been replaced is because LCD displayed produce much better graphics, have better quality and are significantly thinner than the CRT monitors.

For better quality graphics, an LCD screen is used as they come with many features that improve the clarity and speed at which graphics can be seen.

With graphics design, the monitor is much more important when compared to other roles with the computer. The reason for this is due to there being a wide range of different monitors, which means that there are a mix of very high quality colour depth, resolution and screen size. This makes it important to choose the correct screen that is high end to modern standards, which is why I chose the *Dell UltraSharp 34 Curved Monitor U3417W*.

The reason I chose this monitor is because it is a very high DPI and extremely high resolution monitor. It is also a very wide curved monitor that has most of the users point of view so that it can have many windows open for multitasking. The reason I chose Dell for the monitor brand is

due to their famous colour precision, which is loved by graphical artists due to the colours closely resembling the real life version of what the image interprets.

A monitor can be adjusted to suit its hardware surroundings such as scanners, printers and digital devices like the digital camera to produce consistent and precise graphics, which impacts the graphical work in a positive way by stopping deterioration when transferring graphics across different media formats.

Vector based software package

A vector graphic is very different to a bitmap graphic in the may they are both produced and the features they both have. What I mean by this is that a vector graphic is basically an image that is divided into objects, with things such as shapes that have been broken down into lines and curves that can be manipulated via math that the computer does itself.

Vector graphics can have its usefulness when compared to bitmap graphics. For example, vector has a much smaller file size to bitmap graphics, the vector graphic does not pixelate when enlarged, the objects that make up the image have an outline and fill that can be designated by the user and that the objects that are grouped together make editing the image a lot easier.

A vector based software uses mathematical equations and has storage of the properties of an object to create and edit the vector graphic. The mathematical questions are used like commands to determine where each individual object and its properties are set within the vector graphic. The information inside a drawing list which is a list of every object within the image of the vector graphic and it contains the necessary drawing commands used to recreate the vector graphic.

I have picked Adobe's Illustrator application because it is a very good piece of software to create and generate vector images, hence it being the most common application. The software gives the user the ability to create vector graphics for countless uses such as printing, web based graphics, screen based graphics, etc.

The application has multiple features that makes it an effective piece of software. For example, it allows the user to preview lines and shapes before the points are dropped on the 'canvas', it has a "safe mode" which allows it to be reloaded after the application or the Operating System has crashed by saving the file as the user progresses, and it is very compatible with other pieces of software that has been created by Adobe such as Fireworks.

Raster based software package

A raster, or bitmap graphic, is when an image is produced by a large amount of small coloured blocks called pixels. Pixels are assigned a specific colour by the computer and a significant amount of them put together with different colour variations can make a virtual interpretation of an image. However, this does mean that a lot of data will have to be stored in order for the pixels to light up the correct colour. Which, when compared to vector graphics, is a much greater difference in file size.

Raster images have an issue which vector graphics doesn't have, the issue is due to the pixels on the bitmap image being on a fixed position or 'resolution', which causes the image to be 'pixelated', which is when the pixels starts becoming visible to tell from one to another, making the quality of the image decrease significantly.

However, when an image is being scanned, it is easier to convert the scanned image into a bitmap graphic than it is with a vector graphic. The same situation goes for using a bitmap graphic instead of a vector graphic when taking screenshots or using images on webpages. The reason for this is due to the fact that bitmap graphics contain a lot more detail and replicate an image a lot better than a vector graphic would, especially in a web page.

A bitmap editing software is used to edit and manipulate the pixels within the bitmap graphic. What I mean by this is that the software allows the user to change the many aspects of a pixel, such as the colour, the adjustable pixel settings, marge one or more external images, change the colour of the levels, etc. All these features can make an edited and manipulated image look significantly different to the original image.

The reason that I have picked Adobe Photoshop as the software package for creating and editing bitmap graphics is due to the fact that photoshop is also a very well made software from Adobe, which is probably the reason it is the most popular software made by adobe. With this software, many graphics can be developed for a variety of purposes, such as professional or recreational purposes.

Photoshop has many features that aid the user with the editing and creating of images. For example, it has a vast range of file formats, many different collection of filters, the ability to control the levels and colour depth, a collection of simple yet effective tools (i.e. crop, transform, clone, lasso, sponge, etc.)

Online based software application

Just like downloadable software, there are online based applications which can be used for photo manipulation. Just like a traditional photo manipulation software, the online software is a specialised piece of software that can be used to edit bitmap images with various tools via the current browser that is being used.

There are typically a whole suite of tools that a designer can use, each performing specific functions to edit photos and improve the image quality. The online based software application that I have chosen is *https://pixlr.com/editor/*.

The reason I chose this is because it has very similar features to Photoshop, such as similar tools, use of layers, filets, etc. In addition to these features, pixlr allows the designer to use this browser based software on any PC browser that he wishes, instead of downloading software with a large file size.

Task 2

(U30:M1) - Compare the limitations of different hardware and software packages used in

graphics work

Task description: *You are to compare the **limitations** of the following hardware and software packages, ensuring you cover the downfalls of each when compared to the other for the purpose of creating digital graphics.*

A mouse and a graphics tablet



The mouse the the image above is an ergonomic mouse made by Microsoft, called the *Microsoft Sculpt Ergonomic Mouse*. It has what Microsoft claim as the “advanced design” of the mouse which they called the “Sculpt Ergonomic Mouse”, claims to maximize wrist comfort while making navigation “effortless”. The mouse also contains simple navigation with the buttons found on the left of the mouse. However, it is only compatible with a Windows computer.

One of the significant disadvantages of a computer mouse is the location in where it is placed. In order for a computer mouse to work effectively, it needs to have a flat and unobstructed surface.

However, it isn't guaranteed that there will always be a flat surface for the user, especially as more laptops and other mobile devices become much more powerful and compatible with graphics design. Furthermore, the use of mobile devices that are included with trackpads or touchscreen will also make the mouse much more obsolete than it previously was.

Another disadvantage with a computer mouse is that it makes the user prone to wrist or hand injury. The reason for this is due to the repetitive movements of the wrist and hand, which can cause an ache in the tendons on the back of the mand and in the wrist after a prolonged use of time with a computer mouse.

Even though this can be overcome with a mouse that is much more ergonomical and comfortable for the user or if the user’s wrist isn't resting on the desk, there are still bad habits that can be obtained when using a computer mouse which a user doesn't notice.
A Mouse

Since most digital designers spend a lot of time on the computer, one of the most important and most used tools for the design is the mouse. Having flashy features like a gaming mouse is probably what a digital designer would want to have in the mouse, because it would be obsolete

and maybe uncomfortable. It probably is best to find a mouse that is both reliable and ergonomic.



Graphics tablets are designed to make digital artwork easier and more precise than it is with drawing with a mouse. A graphics tablet consists of a flat pad and a specialised electronic pen, graphics tablets are often used by professional digital artists to more accurately place lines and use movements naturally associated with drawing. The graphics tablet above is called the *Wacom Cintiq 22HD.*

The most obvious limitation of a graphics tablet is that it is ill-suited for General Use. What I mean by this is that graphics tablets make a poor choice for basic operations like pointing and clicking to make menu selections.

This is because most interfaces on desktop and laptop computers are designed with a mouse and keyboard in mind, making the operation of a graphics tablet somewhat inefficient.

With a graphics tablet, one of the limitation that it has is that it requires a much larger learning curve than a mouse does. This is because a graphics tablet has a pen to draw with instead of moving the mouse to drag and drop, which means that it is a much harder task to master the tablet.

The reason it’s harder to use the tablet is because it uses a pen to draw an image from scratch or to add onto an image. Everyone usually finds it very hard to draw at first, especially with a graphics tablet because it's a totally new experience

Another drawback of the graphics tablet is the cost of acquiring one. For a digital graphics designer, to obtain a graphics tablet is a costly task, but it also depends on what make and model it is. For an independent artist, this cost can be enough to prevent an investment, even if the system itself can be useful.

<u>An iPad and a laptop</u>



The tablet above is the iPad Pro by apple. With the iPad pro, Apple brought many enhancements in the form of hardware upgrades, such as the 2.25GHz dual-core Apple A9X chip and a 12.9-inch touchscreen with 264 pixels per inch.

However, there is a big limitation that the iPad Pro has, and it's that the iOS limits the potential of the iPad Pro. The iPad Pro runs iOS 9, the same software that powers Apple's considerable smaller iPhones. The multitasking features in iOS 9 make perfect sense for devices with large screen, but they let users see only two apps at a time.

The split View mode in iOS 9 provides a multi-window view, but it simply doesn't do enough to overcome other related shortcomings. The iPad Pro's secondary app switched displays all of the user's Split View-compatible apps in a side column, but it is completely unorganized. The feature only shows three apps at a time, and its large icons don't provide shortcuts to specific functions within apps.



The laptop above is the Macbook Pro made by Apple. With the Macbook Pro, it has a 16:10 aspect ratio, 12-inch, 2304x1440 Retina display at 226 ppi. The panel is LED backlit and boasts in-plane switching (IPS) technology for improved viewing angles. It's also laminated with a new anti-glare coating and sports a new variable refresh rate, making it one of Apple's most advanced display ever,

The Macbook has a dual-core Intel Core M (Skylake Y) processor with integrated Intel HD Graphics 515. Apple lists it as having up to 10 hours of wireless use and 9 hours of video playback. The Macbook has a Force Touch trackpad with support for Force Click and gestures,

allowing for more subtle and powerful interactions. It also has a built-in full-size keyboard with new butterfly switches.

For graphics design, the macbook is very good and was even tailor for graphics design. However, there are certain limitations that the macbook has which the iPad Pro doesn't have. For example, the touch screen on the iPad combined with the new Apple Pen gives the graphics artist the ability to draw on the iPad, creating a new image from scratch. This cannot be done via Macbook Pro because the mac does not have a touch screen in order to carry this out.

Paint and Photoshop

Microsoft's Paint app is a simple graphics application that was installed with all versions of Microsoft's Windows OS. The app gives the ability to create and save files that are usually Windows bitmap, which is a 24-bit file. However, it can also include JPEG, GIF, PNG and a TIFF.

The application has a colour mode and a two-colour black and white, but no grayscale. Due to the simplicity of the app, anyone can learn how to use pain in the matter of minutes, making it the perfect app for someone who is doing graphics for the first time or just want to mess around with the features. It hast has very basic features when compared to in-depth photo manipulation softwares, such as the crop tool, the resize tool, auto shapes, brush, etc.

Photoshop is a software made by Adobe and it is arguably one of the most advanced digital graphic software that can be used for many tasks. Photoshop has a wide range of features and tools that can be used for image manipulation techniques.

For example, with Photoshop you can apply contrast, gradients, layers, masks, brightness, etc. The tools in Photoshop are specifically made and there are different techniques of how to use the tools for photo manipulation and production.

The differences between MS Paint and Adobe Photoshop is that Paint is a simple software for photographic editing, while Photoshop is a software with the same ability but with a lot more variety. One of the significant differences between Paint and Photoshop is the availability of the two pieces of software.

MS Paint is already preinstalled on the computer if you have a Windows OS, which means that you don't have to pay and download for the software, which is what you have to carry out in order to get Photoshop.

Photoshop is far more advanced and more commonly used in the digital graphics industry. However, paint is a completely free and installed program compared to the expensive membership which Photoshop has.

Digital Camera and a Scanner



The camera above is the Sony Alpha A7R II.

The main disadvantage is the storage space. The size of a single digital image can be quite large depending of the megapixel of the camera and it won't be long before the designated storage space on the camera fills up.

This leaves the user with the option of extending the existing storage space with a hard drive or extra memory card. Increase in storage space comes at a price when may seem slightly on the higher side. In certain rare cases, the memory card can become corrupt or can develop other problems which can result in loss of data.

Battery consumption is also a big limitation with digital cameras. Battery consumption is much more faster in case of digital cameras. This makes it necessary to keep a few spare batteries in hand, especially during lengthy photo sessions.

The durability of a camera can also be its biggest limitation. This is why it is vital to pay attention the the ambient weather while using a digital camera as they can be extremely sensitive to heat, extreme cold and moisture in comparison to the old film cameras. A digital camera is susceptible to extreme conditions that may cause malfunction and hence are rendered useless when using in snow, rain or desert areas.

Some digital cameras are very complex. For example, an individual using a digital camera needs to know how to upload and edit images and learn how to do complex tasks when using computers. WIth a wide array of features and functions crammed in a digital camera, for those who aren't technologically inclined may find it overwhelming to learn digital photography.



The scanner above is the Epson Perfection V800 Photo.

Scanned output quality can vary. What I mean by this is that the quality of the scanned output can vary quite a bit, depending on a number of factors. Some of these factors include the quality of the lens inside the scanner, how clean the scanner glass is and the condition of the original

documents. If the original documents are available in electronic format, it is often better to use a program such as Adobe Acrobat to convert them into a PDF format that anyone with Internet access can read.

Scanner maintenance can be expensive. For example, many graphic designers uses high-volume scanners to deal with large quantities of artwork and images. While these high-volume scanners can be good tools, many graphic artists do not realise how costly supplies and scheduled maintenance can be. To keep a high volume scanner running at peak performance, owners will need to replace the lamps regularly and perform maintenance on the camera and also the lens too. This type of maintenance can put on a massive cost, so it is vital that the user of the scanner to weigh these costs against the price of other document imaging solutions.

Scanner are also relatively slow, so graphics artists who own a scanner will need to consider the time involved in scanning all of their artwork and images. Automatic feeders can help to speed the production, however it can still take a great deal of time to scan a stack of documents. In addition, when operating the scanner the user will need to carefully check the scanned images to make sure that all the pages have been imaged. It is not unusual for pages to stick together when going through the automatic document feeder, so the user will need to have knowledge on this and watch out for these problems.

Task 3

(U30:D1) - Evaluate the impact of evolving output mediums on the design and creation of graphic images

Task description: *For each of these it is suggested you do the following:*
- *Introduce the hardware, giving some examples, detailing what the hardware is, what makes it new and different price options.*
- *Discuss what changes the new technology has brought to the world.*
- *Explore any further impacts that the hardware has had on the creation and design of graphic images (e.g things needed – hardware, software, skills etc.)*

4K Monitors (in terms of pixels, hardware to power them, cost, how these images look on other displays)

4K, or Ultra HD, is the next step on the resolution ladder. Just like the HD standard, which covers both 720p and 1080p, Ultra HD covers 4K which is 2160p. The 4K standard covers more than resolution. With 4K, it has also been proposed that the frames per second has had an increase, which means the 4K monitors are getting a better refresh rate over time.

However, for digital graphics, the most important part is the colours of the 4K monitors. WIth Ultra HD, there were major improvements with the colours of the monitors. With 4K having a new colour gamut called Rec.2020 was introduced, which is a much better improvement to the previous Rec.709 colour gamut that full HD had.

The positives of having a 4K monitor for digital graphics is that it can create a much larger workspace on the screen. What I mean by this is that a 4K screen will contain a lot more pixels

than a screen with Full HD or HD, so it has a lot more pixels to work with. With more pixels to work with, the icons and windows on the screen can be smaller in size while still being clearly visible to the user.

This can prove to be very advantageous because it gives a lot more screen space for the user so there is more room for multitasking, which can make the productivity much more efficient than before.

Another positive of a 4K monitor is the quality of the screen. What I mean by this is due to the improved colour depth and the larger amount of pixels, this can make the graphic designer focus on a lot more detail than with a Full HD monitor, which means that a graphic image of better quality can be produced. This can also go with colour quality, because the screen has a much better colour depth, it can see the 32-bit colours which is basically 16.7 million different colours that are visible.

A significant negative of a 4K monitor is the cost of the monitor itself. Since 4K is a relatively new piece of technology, it will cost an extra amount of money to buy a 4K monitor than it is to buy a monitor with Full HD on it.

For example, a 4K monitor has a price range from £400 being the cheapest to £1000 being the more expensive monitors. This could really have an effect on the budget of the graphic designer, which may force him to either get the 4K monitor and sacrifice good hardware for the PC, reducing the performance or to not get the 4K monitor in general for better PC hardware which would improve the performance, but lose the benefits of having a 4K monitor.

Similarly, the performance of the PC can also be affected by the 4K monitor. The reason for this is due to the fact that the 4K monitor has a much larger data requirement than a full HD screen, so a graphics card with “4K ready” would be the best solution to overcome the performance issue. However, with a pre-built PC connected to a 4K monitor, the PC may experience a significant performance reduction due to the vast amount of data that the graphics card has to handle.

File size is also affected when using a 4K monitor. The reason for this is due to the fact that the user will have to use a 4K resolution image in order to see it without pixelation. This would cause file size problems because the user will have to save the file as a 4K image, which would be significantly larger than a Full HD or HD image. However, the user can compress the image to reduce the size of the image file.

<u>VR (different options, motion tracking, limited resolutions, hardware needed, new interfaces and possibilities for graphic artists (tilt brush etc.)).</u>

Virtual Reality, or VR, is basically a computer generated simulation of real life. While Virtual Reality is often talked about as being the future of gaming, it is not yet apparent whether Virtual Reality would actually be the future.

For example, gaming is the most obvious use of Virtual Reality, and it is this industry that will push the development of consumer VR headsets forwards. There are already a limited number of options available to the consumer, including Valve and HTC’s Vive as well as the announced

of the Playstation VR set, which was previously called Project Morpheus.

However, retail businesses, including those in fashion, interior and exterior design, and home refurbishment and decoration industries can also benefit from VR. Manufacturing, design and development roles may also require or benefit from virtual reality design skills.

Given the potential for VR to subvert the traditional viewer experience, it is worthy to consider how the practical benefits of this technology will affect art exhibitions, physical art spaces and just graphical artists in general. WIth VR, a graphical artist could potentially have multiple iterations of the same show in various locations simultaneously.

Despite the many benefits of VR, there are also challenges. Cost and usability are the main limitations to VR, and for some it may be motion sickness. Which can also applied to graphical artist making advertisements. Due to the immersive nature of VR, consumers will be much less forgiving of a bad VR ad than they would of a poorly made TV commercial.

For example, watching badly conceived VR can make the viewer feel sick for the rest of the day. Unlike a TV or photographic advertisement, the graphical artist can't just place a single company logo on a piece of branding for a VR advertisement.

Graphical artists may be showing the expansive power of VR technology and VR will foreseeably have a tremendous impact on the arts, however, some people have their personal preference too. Some may enjoy the traditional canvas or PC to do the work on.

So maybe the turn towards VR in the graphical art is not much different than the uptake of other technologies throughout time. For example, when the tubes of paint replaced hand mixed paint which caused a monumental influence. This scenario will probably also apply to VR, but probably to a greater degree. Not only impacting graphical artists and their work, but also some galleries that hold up the work of the graphical artists and the viewer experiences.

<u>Tablets (differing resolutions, touch displays, trends in design styles due to navigation, portable graphic workstations, limited software)</u>

Tablets are available in different sizes and prices, each coming with different features which will be later mentioned. Some of the more recent tablets that have been getting popular is the Microsoft Surface Pro 4, iPad Air 2, Samsung Galaxy Tab S2 and the Google Pixel C. Each of these tablets each have their own advantages and disadvantages for graphic design. However, most of them in general have similar features.

For example, the size of modern tablets are getting larger with the average size being 4 to 5 inches and 6 to 8 inches. This is ideal for graphical and illustrative users as it allows more surface space for them to move their stylus and arms freely with much more drawing space. With a bigger screen size means that there is a higher amount of pixels so that the screen doesn't appear pixelated, which gives a much larger resolution for the tablets.

One critical feature of a tablet is the inputs and outputs. Tablets often come with a headphone jack and a USB C which is universal and compatible with most computers, this can then allow graphic designers to transfer their work much more easily. Some tablets have docking and

keyboard stations which means it's easier to use for the user as they can type in characters a lot faster than the touchscreen keyboard.

Another important feature that tablets have for graphical work is the pressure sensitivity which some modern tablets have installed within the surface. The pressure sensors most of the tablets have installed have either 256, 512 or 1024 pressure levels, the higher the level of pressure sensitivity the more responsive the tablet will process and gives the user better controlling.

Pressure sensitivity can be controlled by the user's finger or a stylus and normally the user can control the thickness or line and brushed, colour and transparency. For example, if a user is using Adobe Photoshop and drawing an image they can control how large the brush and line is by how hard or how light the touch pressure is.

Most tablets, especially the ones made by Apple, have free cloud storage such as the iCloud. This can be used for backup for important images, sound and video files that need to transfer to other tablets or computers via iTunes.

A graphical designer would create different images whether they be raster or vector on a regular basis and will need to transfer them quickly via other devices so cloud storage can work both ways due to the ability to import and export files. Cloud storage also reduces the inputs needed for the device as the tablet won't need extra wire to transfer the data.

One of the main features that almost all tablets have is a camera. The cameras each have a different range in megapixels, which can go from 4 megapixels to 16 megapixels and above. Tablets usually have a front facing camera and a back facing camera, usually the back facing camera has a larger megapixel count.

The use of the camera can replace the need for a camera, reducing the budget cost for a graphics designer because it would be obsolete to buy a digital camera, unless the user wants a higher megapixel count. With the camera, the user could also take a picture and then edit it via the Photoshop app installed within the tablet (if compatible) and edit it without needing a file transfer.